Fiscal Year:	FY 2012	**Task Last Updated:**	FY 01/18/2012
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	College Park	**State:**	MD
Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: End date extended to 12/31/2013 per PI (Ed., 10/12/2012)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Sunderland, Peter (University of Maryland)		
Grant/Contract No.:	NNX10AD98G		
Performance Goal No.:			
Performance Goal Text:			
	A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station and aboard parabolic aircraft. The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux		

Task Description:	sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. The gas flow rate will be adjusted such that the heat flux to the burner surface matches the heat required to vaporize the appropriate flow rate of liquid or solid fuel. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The flight objectives for BRE are: 1. Burn gaseous fuels in BRE to simulate the burning rates of condensed fuels in a quiescent microgravity environment. Instantaneous flame heat flux will be measured and used to control the rate of fuel supply. Fuels to be simulated include paper, plastic, and alcohols. 2. Burn gaseous fuels in BRE to estimate the ignition and extinction behavior of these condensed fuels in microgravity. 3. Use BRE to estimate the effects of burning at enhanced oxygen conditions in microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	• A second M.S. student joined this project: Yi Zhang. • Michael Bustamante has been selected as a NASA Space Technology Research Fellow for 2011/2012. Bustamante's application was entitled "Burning Rate Emulator Experiments for Spacecraft Fire Safety." Bustamante's fellowship includes a student stipend, faculty advisor funding, on-site allowance, and support for health insurance and tuition and fees. He will conduct research at NASA Glenn and NASA Johnson. • A heat flux gauge was incorporated into the wick apparatus. • An additional array of tests were performed with liquid wicks resulting in incident heat flux data, and flame standoff characterization through the full range of angles. A method was developed for consistent and accurate measurement of the standoff distance. Additional tests were also performed recording burning rate measurements to strengthen the previous data set. • Findings were presented at U.S. National Combustion Meeting (see below). • A basic gas burner was constructed capable of testing at all angles of orientation for the purpose of doing some preliminary testing to assist in the design of the next iteration of the BRE. • Gas burner tests were performed matching the recorded mass loss rates per unit area of methanol during wick testing. Images of the resulting flames were taken, analyzed for flame standoff distance, and compared to the methanol results. • An analytical model of the flames was developed. This predicts flame standoff distance and burning rate for laminar diffusion flames on planar wicks. Gravity can be included in any direction. The predictions are in reasonable agreement with the experiments.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Papers from Meeting Proceedings	Dotson KT, Bustamante MJ, Sunderland PB, Quintiere JG. "Laminar burning on flat wicks at various orientations." 7th U.S. National Combustion Meeting, Atlanta, Georgia, March 20-23, 2011. In: 7th US National Technical Meeting of the Combustion Institute, Georgia Institute of Technology, Atlanta, GA, March 20-23, 2011. Pittsburgh, Pa. : The Combustion Institute, 2011. p. 1741-1745. In: 7th US National Technical Meeting of the Combustion Institute, Georgia Institute of Technology, Atlanta, GA, March 20-23, 2011. Pittsburgh, Pa. : The Combustion Institute, 2011. p. 1729-1735. Printed by Curran Associates, Inc. ISBN: 978-1-61782-780-8 , Mar-2011